

Generation Interconnection System Impact Study Report

For Queue Project Q22-2095

Lincoln 100 kV
September 22, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2095. The Interconnection Customer proposes a 112.0 MW solar photovoltaic facility interconnecting at the Lincoln 100 kV substation in the Illinois area. The request was received on May 25, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 641 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$216,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Joliet / Lake Mirror 100 kV	Joliet / Beltway 100 kV L	109.2	121	14.5
Imperial / Chamberlin 230 kV	Hylebos / Ravenna 500 kV	105.3	1,691	10.5
Canal Road / Newport Tie 230 kV line	Trego / Beaver 500 kV L	101.5	1,848	10.1
Old Forge / Ravenna 230 kV Line	Ravenna / Hylebos 500 kV	100.9	1,620	8.2
Aes Beaver Vly Partners Beaver Valley / Beaver Valley 500 kV L	Cranberry / Beaver Vly 500 kV	100.1	2,786	8.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Joliet / Lake Mirror 100 kV	Reconductor line	\$547,000	9% (\$51,000)
Imperial / Chamberlin 230 kV	Reconductor line	\$8,393,000	1% (\$50,000)
Canal Rd / Newport Tie 230 kV	Reconductor line	\$4,516,000	1% (\$27,000)
Old Forge / Ravenna 230 kV line	Reconductor line	\$5,205,000	1% (\$31,000)
Aes Beaver Vly Partners Beaver Valley / Beaver Valley 500 kV L	Reconductor line	\$9,689,000	1% (\$57,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$216,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.